Security In Windows OS

Comparing Security: Past and Present
Analyzing the Evolution of Security Measures

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Introduction

Windows is one of the most widely used operating systems in the world.

Since its first release, it has undergone significant evolution, particularly in security, adding features to protect users from increasing cyber threats. However, it has also been more vulnerable to malware than other platforms. **In this presentation**, we will explore the development of Windows security from the earliest versions to Windows 10 and beyond, highlighting key improvements and the future direction of OS security.

Windows Security Evolution 1985-2000

Windows 1.0 (1985)
– No security features

Windows 2.0 (1987)
– No security improvements

Windows 3.0 (1990)
– Introduced memory protection (Protected Mode)

Windows 3.1 (1992)
– Optional password lock in corporate editions

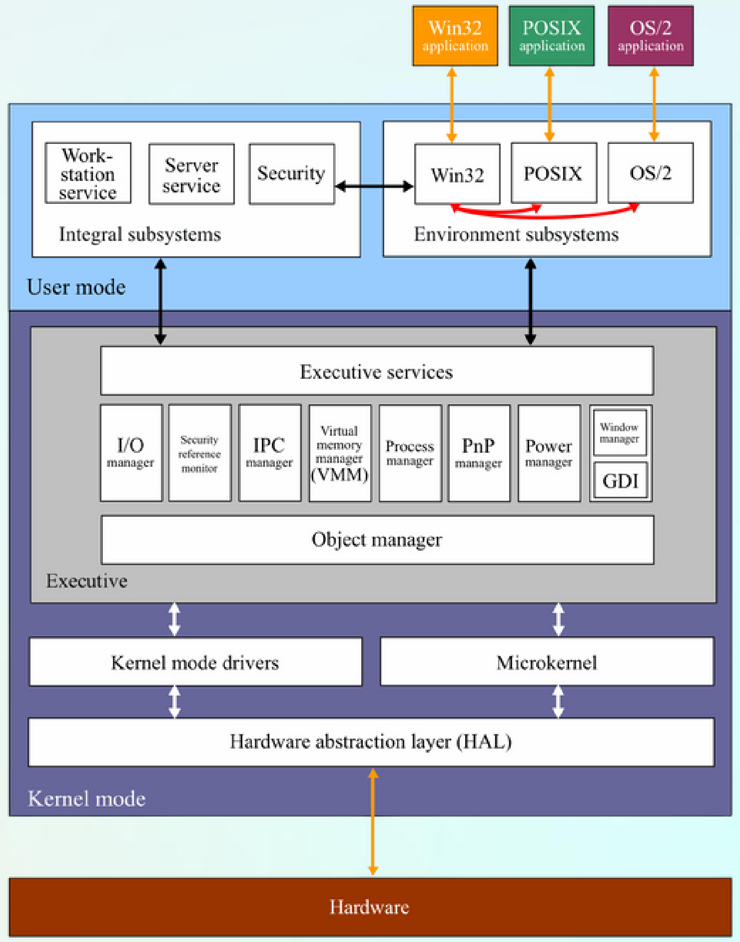
Windows NT 3.1 (1993)
-Multi-user support
-NTFS with ACLs (access control)
-User mode / Kernel mode separation

Windows 95 (1995)
– Very limited security, basic login

Windows NT 4.0 (1996)
-Domain login support
-Security groups and clear user accounts (Admin/User/Guest)

Windows 98 (1998)
– Online security updates introduced

Windows 2000 (1999/2000)



- Active Directory for user & policy management
- Kerberos authentication
- Encrypting File System (EFS)
- Initial firewall and regular updates

Windows XP

The next major OS security facelift fo Windows came with Windows XP in 2001. Sold as Microsoft's most secure OS ever, Windows XP ended up becoming the most patched because of its widespread use. Among the security improvements in Windows XP were

- AutoPlay: Allows the OS to identify when removable media was inserted
- Improved encryption capabilities
- Improved DPAPI security by using a SHA1 hash of the Master Key Password
- Password Reset Wizard: Uses a password reset disk
- Credential manager: Stores user credentials for user accounts
- Introduction of Windows Security Center: Continuously monitors security and services

Windows Vista vs Windows 7 Security Features

Feature	Windows Vista	Windows 7
User Account Condetrol (UAC)	Introduced. Prevents unauthorized system changes	Improved.Less intrusive with smarter prompts
Windows Defender	Built-in anti-spyware Protects against basic spyware	Improved with additional features including support for removable drives and stronger encryption
BitLocker	Disk encryption available	Enhanced. Protects more effectively against spyware and unwanted software
Data Execution Prevention (DEP)	Partial support. Stops some malicious code execution	Full support Prevents execution of malicious code in protected memory regions
Address Space Layout Randomization (ASLR)	Not widely used	Implemented Randomizes memory addresses to prevent memory-based attacks

Windows 8 and Windnws 8.1

A New Era of Security

Following Windows 7, Microsoft released Windows 8 and 8.1 (2012–2013) with **enhanced security to shield users from contemporary threats.** With the introduction of the Start screen, live tiles, and full touchscreen support, Windows 8 signaled a significant change in design. Many users had trouble adjusting, so Windows 8.1 improved the experience by maintaining the contemporary appearance while enhancing navigation and usability for both desktop and tablet users.

Windows 8 (2012)

- **User Account Control:** prevents malware and unintentional changes to your computer by temporarily blocking changes until an administrator gives their approval.
- **Windows Defender:** keeps your computer safe from malware and viruses while providing real-time file and application scanning without causing system lag.
- **Windows Firewall**: protects your computer from network threats and blocks unwanted access to your Internet connection.

Windows 8.1 (2013)

- **Trusted Platform Module (TPM 2.0):** TPM 2.0 protects cryptographic keys and makes sure the integrity of devices.
- **SecureBoot:** prevents malicious software from loading before Windows boots up.
- **Data Protection (Encryption):** Automatic encryption and remote wipe secure corporate data on personal devices
- **Windows Defender:** Enhanced Windows Defender detects malware behavior before signatures exist

Windows 10 security features (2015)

Windows 10 is the newest version of Microsoft's operating system. It is effective on PCs and tablets while providing **enhanced protection against cyber attacks** and invasion of privacy. Its strong security capabilities enable businesses to safeguard networks and confidential information, equipping cybersecurity teams with advanced tools to avoid and respond to looming threats

- **Windows Defender Antivirus (WDA):** Antivirus with real-time protection, firewall, and safe browsing all built in.
- **Microsoft SmartScreen:** Blocks malicious programs and warns against suspicious websites and emails
- **Windows Defender Application Guard:** Isolates untrusted websites to prevent threats from reaching your system
- **Windows Sandbox:** Runs new apps in isolated virtual environments to prevent full threat exposure.
- **User Account Control (UAC):** Prevents unauthorized changes by asking for admin permission
- **Microsoft Defender Advanced Threat Protection (ATP):** Monitors endpoints with behavioral sensors and cloud analytics

Windows 11 Currently

Windows 11 is Microsoft's **most secure version yet,** built on a modern security architecture that combines hardware-based protection and cloud intelligence. Reducing vulnerabilities from the first startup.
According to Microsoft, organizations saw a 58 % drop in security incidents after upgrading from Windows 10.

Advanced Built-in Modern Protection

- **Smart App Control:** A new feature using cloud-based AI to predict app safety, blocking untrusted or malicious software automatically. Requires a clean installation of Windows 11 to be fully enabled.
- **Virtualization-Based Security (VBS):** Enabled by default on new PCs, VBS isolates critical system memory to protect credentials and security functions.

Data and Threat Defense

Security tools such as **BitLocker** and Microsoft Defender **Antivirus** protect data through full-disk encryption and real-time malware detection. These integrated features reduce the need for external antivirus software and automatically handle most common threats.

Current Challenges and Advanced Tools

Windows 11 still faces evolving risks like **ransomware** and new vulnerabilities, but Microsoft responds quickly with regular security updates. Additional tools such as Windows **Sandbox** and **Controlled Folder Access** allow users to test untrusted apps safely and keep critical files protected.

AI in Windows Security

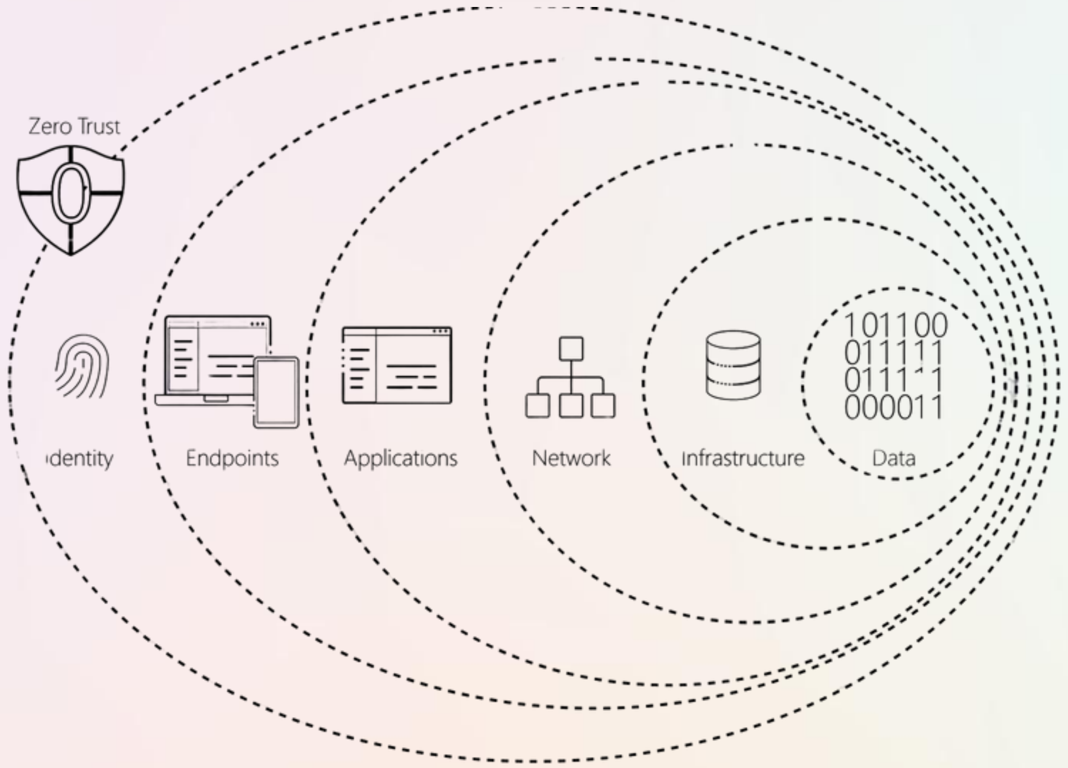
- Uses Defender SmartScreen, Defender XDR, Security Copilot with cloud-based AI.
- AI detects suspicious activity and blocks attacks automatically
- Microsoft announced that future Windows releases will rely even more on AI-powered anomaly detection and automated response systems.

Copilot

Your everyday AI companion

Zero Trust and Cloud Security

- Continuous verification of all users and devices
- Future cloud AI tools will predict global threats and coordinate automatic responses across devices



Persistent Threats and Future Solutions

- Advanced ransomware, zero-day exploits, and phishing are still challenging today.
- Future solutions: AI-driven detection, post-quantum encryption, and self-healing systems to stop attacks before damage occurs.

into the Future Security

THANK YOU!

References: **Click it!**